Project Uppercut — First Bell readiness

The number that matters is not how many tests pass. It is how many fights the integrated contract has actually run from bell to decision.

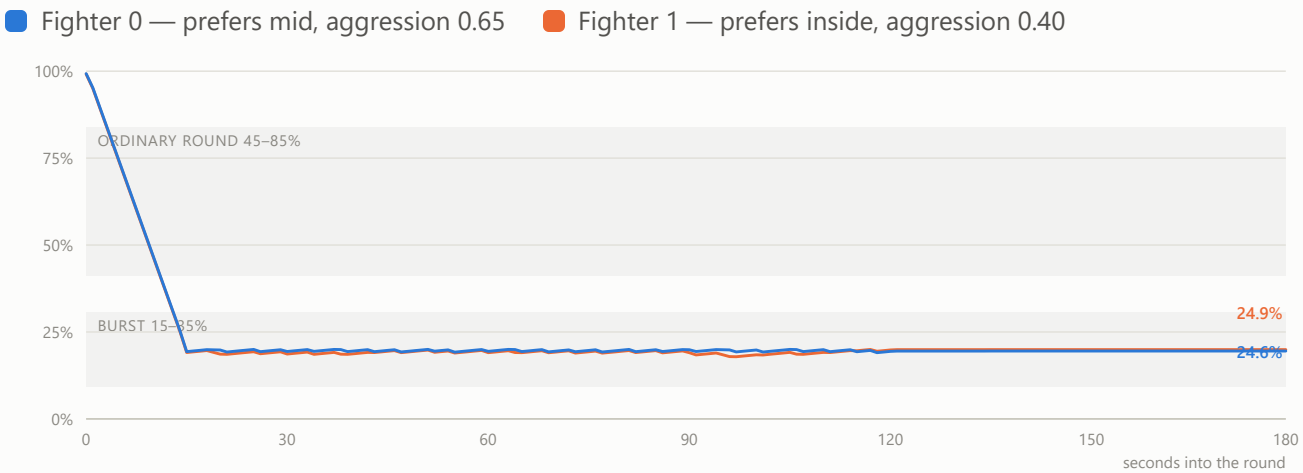
Branch `fix/one-wedge-killer` · head `cc8bc5a` · PR #23 green and mergeable · generated 2026-08-08 from a single diagnostic round, seed 4242

0 complete integrated fights

The contract, the named RNG substreams and the observable CPU policy all exist and are green. Nothing has yet started at the bell, gone to a stoppage or a decision, and been sealed. Until one has, four of Decision 0022's seven required assertions cannot be tested at all — and they are named in the suite rather than replaced by test-shaped substitutes.

THE FIRST ROUND THE POLICY EVER RAN

One 180-second round, both fighters driven by the new observable policy, both charged by the proven Wind/Reserve economy. The shaded bands are the acceptance targets `stamina_economy.json` already declares — unfitted hypotheses, not tuned values. Neither fighter enters the ordinary band at any point after the first 12 seconds. The two traces coincide from about 15 seconds on — they end 0.4 points apart, which is why both carry a direct label and the table view is here.



▶ Table view — Wind %, sampled once per second

WHERE THE WIND ACTUALLY WENT

Fighter 0's round, attributed by action family: how many ticks chose it, and the signed Wind the economy applied on those ticks. The ledger closes to the reported end state, which is the only reason the chart above can be read as a finding rather than a shape.

ACTION FAMILY	TICKS	NET WIND	NET	PER TICK
move	1,004		-84.34	-0.08400
attack	33		-19.64	-0.59515
observe	429		+1.46	+0.00340
hold_position	9,334		+27.07	+0.00290

1,004 ticks of footwork cost more than three times what 9,334 ticks of holding position paid back. `rest`, the only state that pays well, does not appear: it was never chosen.

WHAT THIS RUN FOUND

Three findings, in the order they would bite. None of them is tuned here — ruling R5 holds until the policy produces plausible cadence, and the canon values are the Director's call.

F1 The units error, a third time - and this one was mine

The probe read `locomotion.modes[0].stride_length_ru` (0.55, free_navigation) and charged the economy that distance every tick. A stride is the length of a step; canon declares `max_frame_displacement_ru = 0.14` as "a hard per-tick displacement cap below every stride". The probe was moving a fighter 3.9× faster than canon permits and billing him for all of it. Fixed before any number here was quoted - every figure on this page is from the corrected run.

The repo's units guard is one-sided. `economy_pack` multiplies every recovery rate by the round length and fails if one could refill the pool. Nothing multiplies a drain rate by the round length, so a per-step distance billed per tick passed every gate in the repository.

F2 Rest was chosen zero times in a 180-second round

Recovery was the decision on 9,317 of 10,800 ticks, and not one of them was `rest`. Policy branch 7 rests only at `outside` / `out_of_range` and holds position otherwise - correct

boxing, since you do not stand and breathe with a man in front of you. But no branch anywhere converts *I need air* into *make room*. There is a reason code for leaving a bad range and none for leaving to breathe.

Consequence: every recovery tick paid `hold_position`'s 0.0030/tick instead of `rest`'s 0.0060 - exactly half. Caveat held open: the probe's range model is a declared stand-in in which both fighters share one band, which makes separation harder than a real range subsystem would. The missing branch is real either way; its magnitude here is not evidence.

F3 Movement costs 14 seconds of rest per second of footwork

At the declared cap, footwork costs 0.084 Wind/tick against rest's best 0.0059 - a ratio of 14:1. A fighter who rested every tick of a round would bank 63.7 Wind; with a full starting pool that is about 32 seconds of footwork inside a 180-second round. Both fighters threw 33 punches.

Not tuned, and deliberately so. Ruling R5 forbids stamina tuning until the observable policy produces plausible cadence, and canon values are the Director's call. This is the first evidence that could inform that call rather than a request to make it.

DECISION 0022 — THE SEVEN REQUIRED ASSERTIONS

Three have real coverage. Four cannot be tested until their subsystems exist, and the suite contains a test whose only job is to keep them named. Naming an unavailable proof beats inventing a test-shaped substitute for it.

#	ASSERTION	STATE
1	Same seed and initial state reproduce byte-identically Header layer only. The bout has no event rows yet, so nothing below the header is being compared.	• Covered
2	Different seeds create non-vacuous variation Including that the substreams themselves differ, not merely the recorded seed line.	• Covered
3	All event families appear in one stream There are no event families yet. Naming this is the point.	○ Not proven
4	No system reads another subsystem's private draw Authority is declared per stream and visible in the artifact; the structural half is done, the behavioural half needs subsystems.	◐ Partial
5	Pause, speed and camera do not alter authority Presentation has its own stream, but nothing presentational runs yet.	○ Not proven

#	ASSERTION	STATE
6	One runtime serves commandlet and live arena The convergence rule's own gate. Waits on the integrated runtime.	○ Not proven
7	A bout ends by stoppage or decision Nothing can end a bout yet. This is the last one to fall.	○ Not proven

ENGINE PHASES — RE-PROVED, NOT REMEMBERED

Every frozen phase now has a prover that rebuilds the C++ artifact and compares its marker stream byte for byte against the analyzer's. Six phases had no prover before this sprint. The first full run found that one of them had silently drifted.

PHASE	WHAT IT PROVES	STATE
H	The C++ core scaffold 2 seeds x 600 ticks	• Proven
I	The two-fighter authoritative loop 2 seeds x 1200 ticks	• Proven
J	Real movement and swept collision 2 seeds x 1200 ticks	• Proven
K	The first rendered slice 1 seed x 5400 ticks - repaired, reproduces the pinned hash 274c597b	• Proven
L	Guard and slip 2 seeds x 1200 ticks	• Proven
N	Visible stamina and condition 2 seeds x 1200 ticks - had drifted; found by the new prover and repaired	• Proven
Q	Core range Own prover - wedge killer took it 13 min to 50 s	• Proven
R	The head is not where the punch is going Own prover	• Proven
O	Contract only No C++ artifact exists to re-prove	– N/A
INT	The integrated bout Contract, RNG and policy exist. No bout has been run end to end.	○ Not proven

THE ONE DESIGN QUESTION THIS RAISED

Decision 0022 names RNG substreams by subsystem, not by fighter. With two fighters both drawing from `policy_intent`, fighter A's decision count shifts fighter B's draws — the same class of coupling 0022 forbids *between* subsystems, reappearing *within* one. This probe used a fixed fighter-0-then-fighter-1 order as a declared diagnostic choice. The integrated bout will need a ruling: per-subsystem streams, or per-subsystem-per-fighter.

Raising it now, while the only cost is a naming rule, is cheaper than discovering it when a knockdown at tick 8421 moves because the other man threw one more jab.

A harness may explore. Only the contract may conclude. This page is a harness: nothing on it is pinned evidence, no hash was taken, no canon value changed, and `values_are_fitted` remains false everywhere.